

Chaitanya Murarka

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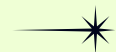
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Summary



Full Stack & Quant Engineer specializing in real-time trading platforms, low-latency data pipelines, and distributed backend systems. Architected and scaled an end-to-end algorithmic trading platform across FastAPI microservices and React/Next.js micro-frontends, with a strong foundation in data science and ML.

Experience



Secret Weapons Trading Solutions Pvt. Ltd.

Software Engineer

Remote | Jun 2025 – Current

- Architected and shipped a production algorithmic trading platform as 9+ FastAPI microservices behind an API gateway, with a React/Next.js micro-frontend and Keycloak SSO serving live multi-user trading.
- Engineered real-time market-data ingestion pipelines (DTN IQFeed tick and Level-2 depth → Redis Streams → TimescaleDB) with sharded, fault-tolerant processing and auto-reconnect, reducing ingestion CPU from ~73% to ~9%.
- Designed and delivered 15+ quantitative trading strategies and indicators with backtest-to-live parity, multi-timeframe gating, and automated live execution via broker webhooks.
- Improved performance by orders of magnitude through JIT compilation (~150x), vectorized computation (~700x), and front-end rendering optimizations (~84% lower CPU) for smooth real-time charts under load.
- Migrated the platform's time-series data layer to TimescaleDB and re-architected the strategy engine for improved maintainability and horizontal scalability across services.
- Integrated subscription billing and access management with Outseta and Stripe, covering authentication, entitlement gating, and self-serve plan management.
- Owned end-to-end delivery of TickPro (Electron multi-engine desktop analytics app) and Trade Distributor (cross-broker copy-trading — Zerodha, Shoonya/Finvasia — with per-order latency tracking).
- Built and deployed a standalone live tick-recording service (Redis stream to daily S3 session datasets) with automated email delivery and a Keycloak-gated download dashboard at data.marketforecaster.com.
- Managed AWS infrastructure and CI/CD, cutting cloud cost ~55% (~\$67 to ~\$30/day) and mentoring junior developers via code reviews, hiring, and task planning.

Algoji Enterprises Pvt. Ltd.

Strategy Developer Intern

Remote | Jan 2024 - Jun 2024

- Re-engineered the API Bridge trading software's internal master symbol code with multiprocessing, significantly reducing generation latency under high volume.
- Built real-time data feeds into AmiBroker from broker APIs (Finvasia/Shoonya, Symphony XTS) via an ODBC/MySQL pipeline with NSE/NFO/BSE/MCX scrip-master mapping.
- Developed a Python XTS charting/trade-tracking app and live-quote-to-Excel feeds, plus an auto-updater for broker master-contract files (Windows executables).
- Automated derivatives execution (option chain, Greeks, basket and SPAN margin) and TradingView-to-API Bridge order routing across brokers; authored Pine Script strategies.

Solar Industries India Ltd.

Data Analyst Intern

Nagpur | Oct 2022 - Apr 2023

- Drove supply-chain and production analytics with time-series demand forecasting and Python/Tableau dashboards for executive insights.
- Led a computer-vision project to monitor production output and stock levels and enforce compliance with safety regulations, annotating datasets with LabelImg and training YOLO object-detection models.

Education

2020 to 2024

Computer Science & Engineering (Specialization in Artificial Intelligence and Machine Learning)

B.Tech, Shri Ramdeobaba College of Engineering and Management, Nagpur

Independent Projects

[AlertSync](#) — Full-Stack Alert Automation SaaS

- Built a TradingView-to-Telegram alert-relay SaaS (FastAPI, Next.js, Keycloak) with multi-tenant, Redis-backed quota and state management.
- Integrated Razorpay billing with dual one-time and auto-renew subscription checkout, HMAC signature verification, and idempotent webhook handling.

[PDF2.in](#) — Browser-First PDF Toolkit

- Built 20+ privacy-first PDF tools (merge, split, compress, redact, watermark, sign, OCR, format conversion) in a Next.js 14 app — processing files client-side via pdf-lib/pdf.js, with a multi-mode Sign-PDF tool and a lightweight FastAPI backend deployed via Docker/CI-CD.

[Volatility Breakout Screener](#)

- Real-time WebSocket screener with multi-timeframe Bollinger Bands + ATR breakouts.

Technical Skills

- **Backend & Infrastructure:** Python, FastAPI, REST APIs, Microservices, Redis (Streams & Pub/Sub), PostgreSQL, TimescaleDB, InfluxDB, Docker, Keycloak, NGINX, Linux
- **Frontend:** Next.js, React, TypeScript, JavaScript, Electron, Socket.IO, WebSockets, Lightweight Charts, Event-driven Micro-frontends
- **Cloud & DevOps:** AWS (EC2, ElastiCache, IAM), Cloudflare, GitHub Actions, CI/CD, Certbot/TLS, fail2ban/UFW, Cost Optimization
- **Quant & Trading Systems:** Strategy Development, Backtesting, Technical Indicators, Real-time Market Data Pipelines, Numba JIT, Vectorization, Performance Profiling
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, Time Series Forecasting, Statistical Modeling, Feature Engineering, EDA, Data Visualization

Certifications and Licenses

- AWS Fundamentals
- NoSQL & Big Data Foundations